Precision Subtyping of Dementia Through Integrative Multi Omic Modelling

PREREGISTRATION POSTER

Alieyeh Sarabandi Moghaddam [1,3], Esteban Gomez Cifuentes [2], Emma Squires [1,3], Alfredo Iacoangeli [2]

1 Swansea University Prifysgol Abertawe
2 King's College London
3 Dementias Platform UK | Data Portal

Introduction

Dementia consists of multiple neurodegenerative diseases with overlapping clinical features and heterogeneous molecular drivers. Traditional diagnostic labels fail to capture this biological diversity, limiting prognostic accuracy and treatment stratification.

We propose a precision subtyping framework that integrates multi-omic datasets to identify mechanistically grounded dementia subtypes that transcend current clinical categories.

Materials

Multi-omic data from Rush Alzheimer's Disease Center cohorts, including genotyping, mRNA, epigenetic, proteomic, metabolomic, and ***autopsy-derived neuropathology***, will serve as the primary analytic resource. The data is spread across multiple cohorts with distinct racial and ethic backgrounds, and expansion is ongoing (ROS, MAP, MARS, AA core, LATC).

Genotyping
Whole genome sequencing data

mRNA
snRNAseq PFC, RNAseq PFC microglia, PFC microglia

Proteomic
Proteomics Somalogic, Proteomics TMT

Metabolomic
Brain lipidomics, Brain metabolomics, Serum lipidomics, Serum metabolomics, Bile Acid

Epigenetic
ATAC Seq, H3K9Ac,

Analysis of five harmonized longitudinal cohorts includes over 4,600 participants for dementia and over 3,900 for mild cognitive impairment (MCI). Omics and autopsy data is more limited in size.

Methods

Standardized preprocessing includes
- modality-specific quality control (e.g., MAF >1%, RIN >7, proteomics normalization),
- ComBat batch correction,
- multi-omic factor integration via MOFA+.

Subtype discovery will incorporate dimensionality reduction (PCA, UMAP) followed by semi-supervised clustering constrained by established **genetic variants, Braak staging, and cognitive impairment severity**.

Subtype distinctiveness will be assessed via **silhouette score** (>0.65) **and cluster stability**. Pathway significance will be evaluated at FDR <0.01 using **over-representation and gene-set enrichment analyses.** Regulatory network inference (NicheNet, WGCNA) will integrate snRNAseq with ATAC-seq and H3K9Ac to link **epigenetic states to transcriptional drivers.**

Biological validity will be tested through **correlations with quantitative neuropathology and longitudinal cognitive decline**.
External reproducibility will be examined via resampling and sensitivity analyses to confirm robustness to batch structure and data imbalance.

Model Layers

Multi-omic integration will be performed using MOFA+ on a core set of data layers:

- **Phenotypic anchors** will be whole-genome sequencing (for polygenic risk scores and rare variant constraints), snRNAseq from PFC (for cell-state resolution), Somalogic and TMT proteomics (for systemic and brain-specific protein readouts), and autopsy-derived neuropathology (Braak, CERAD, TDP-43, vascular pathology).
- **Supporting layers** including PFC microglia RNAseq, ATAC-seq, H3K9Ac
- All brain and serum metabolomics/lipidomics will be used for **downstream mechanistic deep-dives and translational biomarker identification**, not as primary clustering inputs.

Expected Outcomes: Aims & Hypotheses

Aim 1: Identify multi-omic dementia subtypes via unsupervised integration of genome sequencing, snRNAseq, proteomics, and neuropathology.

- Hypothesis 1: Integration of omic layers will resolve 3–5 stable, reproducible subtypes that are not captured by clinical diagnosis alone and exhibit distinct molecular and pathological signatures.

Aim 2: Characterize cell-type-specific regulatory mechanisms underlying each subtype.

- Hypothesis 2: Each subtype will be defined by distinct microglial activation states, neuronal vulnerability patterns, and epigenomic regulatory programs (ATAC-seq, H3K9Ac) that drive downstream transcriptomic and proteomic alterations.

Aim 3: Translate brain-derived subtypes into blood-based biomarker signatures.

- Hypothesis 3: Serum proteomics (Somalogic) and metabolomics/lipidomics will contain discriminative features that accurately classify brain-defined subtypes, enabling non-invasive stratification and prognostic monitoring.

Aim 4: Validate subtype specificity and clinical relevance.

- Hypothesis 4: Subtypes will demonstrate differential rates of cognitive decline, distinct neuropathological burdens, and reproducible assignment across resampled cohorts, supporting biological validity and utility for precision medicine.

Impact & Future Steps

This work will reframe dementia not just as a collection of clinical labels but as a spectrum of convergently disrupted biological systems, with immediate impact on future clinical trial design by enabling prognostic enrichment as well as mechanism-matched therapeutic targeting.

Next steps include:
- External validation in independent autopsy cohorts,
- Prospective testing of serum biomarker panels
- Functional validation of candidate drivers in iPSC-derived microglial and neuronal models to establish causality and prioritize drug targets

Given the limited overlap of omics data in currently available datasets, undergoing the analysis again once more data is available is Crucial.